



What is Indexing?

Indexing means **accessing a single element** from a string or list.



Character	P	Y	T	H	O	N
Index	0	1	2	3	4	5
Negative Index	-6	-5	-4	-3	-2	-1

✓ Example:

```
name = "PYTHON"
print(name[0]) # P
print(name[3]) # H
print(name[-1]) # N (last character)
print(name[-3]) # H
```



What is Slicing?

Slicing means **taking a part (sub-string or sub-list)** from a string or list.

Format:

```
variable[start : stop]
```

start

index where slice begins (inclusive)

stop

index where slice ends (exclusive, not included)



Basic Slicing Example



```
name = "PYTHON"  
print(name[1:4])
```

❏ **Output:** YTH (Because characters at index 1,2,3 are included → 4 is not included)

★ Useful Slicing Examples

1

From beginning to some index

```
name = "PYTHON"  
print(name[:3]) # Output: PYT
```

2

From some index to end

```
print(name[2:]) # Output: THON
```

3

Full string copy

```
print(name[:]) # Output: PYTHON
```

4

Using negative index

```
print(name[-4:-1]) # Output: THO
```



Step Slicing

Format:

```
variable[start : stop : step]
```

Example:

```
name = "PYTHON"  
print(name[::-2])  
# P T O (every 2nd character)  
  
print(name[::-1])  
# Reverse the string → NOHTYP
```





Examples with List

```
nums = [10, 20, 30, 40, 50]

print(nums[0]) # 10
print(nums[-1]) # 50
print(nums[1:4]) # [20, 30, 40]
print(nums[:3]) # [10, 20, 30]
print(nums[2:]) # [30, 40, 50]
print(nums[::-1]) # [50, 40, 30, 20, 10] (reverse list)
```



Quick Summary

Concept	Meaning	Example	Result
Indexing	Access 1 value	<code>name[2]</code>	T
Negative Indexing	From end	<code>name[-1]</code>	N
Slicing	Get part of string	<code>name[1:4]</code>	YTH
Step Slicing	Skip characters	<code>name[::2]</code>	PTO
Reverse	Reverse sequence	<code>name[::-1]</code>	NOHTYP